

Executive Summary

This report shows the findings of a Monte Carlo simulation comparing two portfolios over a 10-year investment horizon with an initial investment of \$100,000. It compares a pure gold investment against a traditional diversified 60/40 multi-asset portfolio.

The results showed that the diversified portfolio is superior, with a higher median and lower downside projection. While gold is projected to have greater potential in favourable environments, its higher volatility leads to a wider range in outcomes. These findings show the long-term benefits of diversification for most investors with a standard risk tolerance.

Monte Carlo Simulation Methodology

Monte Carlo simulation is a quantitative technique that uses repeated random sampling to model uncertainty and probability distributions of potential future outcomes. In this study, 10,000 iterations were conducted for each portfolio. Annual returns were drawn from a normal distribution using each strategy's expected return and volatility parameters. The terminal wealth values were calculated by compounding the annual returns over the 10 years.

This approach provides a way to evaluate both the expected returns and potential outcomes, including tail risks. This offers a deeper insight than other forecasting methods.

Portfolio Descriptions and Strategic Rationale

Portfolio A – 100% Gold Allocation

This strategy remains fully invested in gold, an asset historically valued for safeguarding capital and reducing risk associated with inflation, currency depreciation, and geopolitical uncertainty. Gold is typically shown to have a low correlation with traditional financial assets, making it an effective and attractive diversifier in certain economic markets.

Portfolio B – Diversified Multi-Asset Portfolio

This portfolio represents a balanced allocation, distributing capital across a 60/40 equity-to-fixed-income ratio. This strategy optimises the risk-return frontier by capturing the equity risk premium for long-term growth while utilising bonds to reduce volatility and provide income stability. A diversification strategy can successfully generate superior risk-adjusted returns across fluctuating economic cycles.

The comparison between these approaches allows for a clear evaluation of concentrated risk versus the benefits of broad asset class exposure.

Key Assumptions and Parameter Selection

The simulation includes the following parameters, selected based on long-term historical data and current strategic capital market assumptions:

- **Gold Portfolio:** Expected annual return of 7.0% with annualised volatility of 18%.
- **Diversified Portfolio:** Expected annual return of 8.0% with annualised volatility of 12%.

These figures are taken from practical research. Gold's return and volatility estimates reflect its performance over multiple decades. This includes periods of high inflation and economic uncertainty. The parameters used for the diversified portfolio are consistent with observed results for diversified portfolio strategies and current market forecasts from leading institutions.

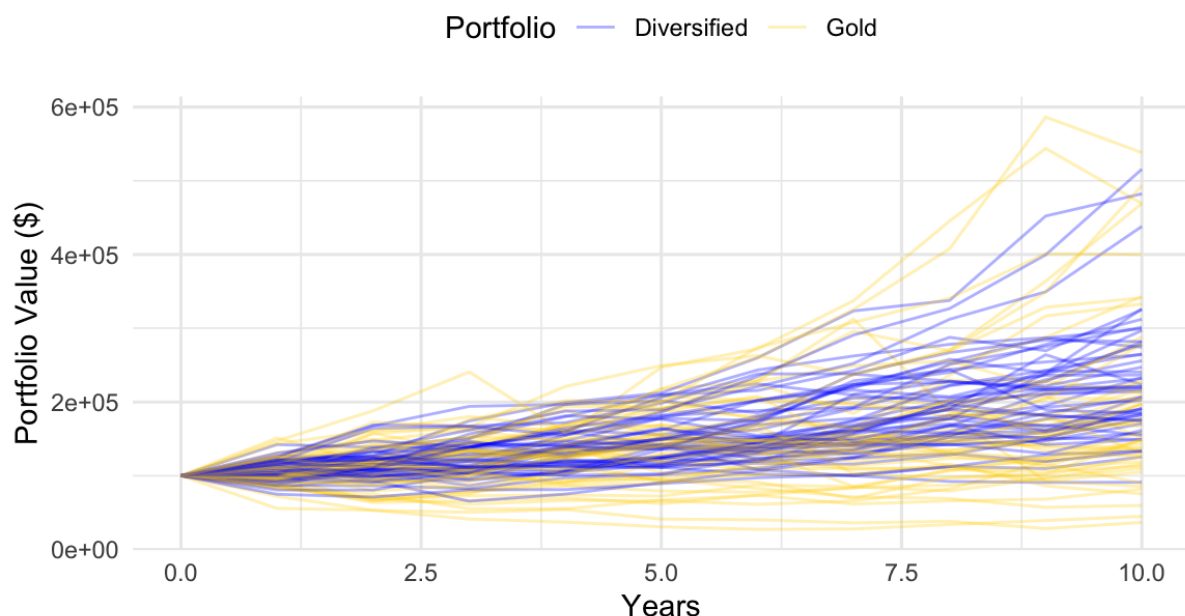
For simplicity, this analysis assumes normally distributed annual returns, no transaction costs, taxes, or rebalancing fees. There is also a focus on nominal returns rather than inflation-adjusted returns. These assumptions allow for a direct and simplified comparison while maintaining clarity.

Simulation Results and Analysis

50 Random Trial Paths

The graph below shows 50 randomly selected wealth trajectories for both portfolios over the 10 years, with the blue lines representing the diversified portfolio and the gold lines representing the gold portfolio.

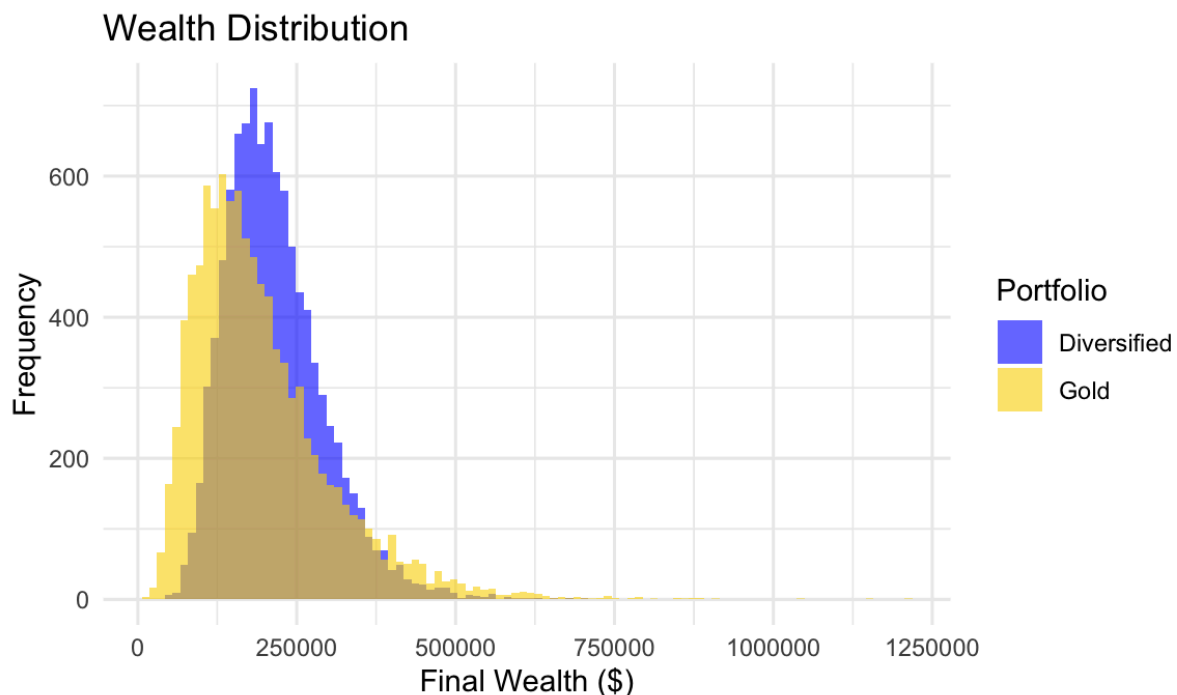
Trial Paths for Both Portfolios



The graph shows higher variability/volatility in the gold portfolio, with several lines showing improved gains alongside others with considerable losses. The diversified portfolio shows more consistent growth patterns with narrower variability. This shows that gold is more sensitive to market conditions than the diversification of the portfolio.

Ending Wealth Distribution

The graph below shows the distribution of the final value/wealth values across all 10,000 simulations, with the blue bars representing the diversified portfolio and the gold bars representing the gold portfolio.



The diversified portfolio shows a tighter distribution with its mean towards the right of the gold portfolio's mean; this indicates a higher probability of the diversified portfolio achieving a more favourable outcome. The gold portfolio's distribution is wider, indicating more uncertainty and likelihood towards extreme outcomes.

Percentile Analysis

The percentile results offer a look into the potential outcomes:

- In the 10th percentile (worst-case scenarios), the gold portfolio ends at approximately \$81,959, while the diversified portfolio reaches \$127,429.
- At the 50th percentile (median), the diversified portfolio achieves \$204,279 compared to \$170,288 for gold.
- In the 90th percentile (best-case scenarios), the gold portfolio shows a strong performance at \$341,110, while the diversified portfolio reaches \$317,830.

These results show gold's asymmetric and volatile characteristics with higher potential returns in strong environments but significantly increased downside risks.

Risk Assessment

The 95% Value-at-Risk (VaR) analysis reinforces the diversified portfolio's defensive characteristics. The gold portfolio faces a higher probability of losses in adverse market conditions, whereas the diversified portfolio shows greater resilience, keeping the worst-case outcome near or above the initial investment.

Conclusion and Investment Implications

The Monte Carlo simulation results strongly suggest diversification over a concentrated gold allocation for long-term investors. While gold serves an important role in portfolio construction, a 100% allocation introduces volatility and downside risk relative to the more balanced risk-return profile offered by a diversified multi-asset strategy.

Investors should consider incorporating gold within a broader, well-diversified portfolio rather than relying on it as a standalone solution. The findings align with established portfolio theory emphasising the benefits of asset class diversification for optimising long-term wealth accumulation while managing risk.

Sources:

JP Morgan. "2026 Long-Term Capital Market Assumptions." *JP Morgan*, 2026,

UBS. "Global Investment Returns Yearbook 2026." 2026.

World Gold Council. "Gold Data | Gold Statistics | Goldhub." 2026.